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Instructions Follow instructions *carefully*; failure to do so will result in points being deducted.

1. [16 points] Using the **formal definition of Big-O**, show that each of these functions is $O(x^2)$. You must clearly state what the witnesses to the relationship are to receive full credit.
   (a) $f(x) = 26x$
   (b) $f(x) = 20x + 26$
   (c) $f(x) = 20x^2 + 2x + 6$
   (d) $f(x) = 20x^2 + 26log(x)$

Solution: Enter your solution here:
A:
For $x \geq 1$ we have $x \leq x^2$, so

$$26x \ \leq \ 26x^2.$$

Thus $f(x) \in O(x^2)$ with witnesses $c = 26$, $k = 1$.

B:
For $x \geq 1$ we have $x \leq x^2$ and $1 \leq x^2$, so

$$20x + 26 \ \leq \ 20x^2 + 26x^2 \ = \ 46x^2.$$

Thus $f(x) \in O(x^2)$ with witnesses $c = 46$, $k = 1$.

C:
For $x \geq 1$ we have $x \leq x^2$ and $1 \leq x^2$, so

$$20x^2 + 2x + 6 \ \leq \ 20x^2 + 2x^2 + 6x^2 \ = \ 28x^2.$$

Thus $f(x) \in O(x^2)$ with witnesses $c = 28$, $k = 1$.

D:
For $x \geq 1$ we have $\log(x) \leq x \leq x^2$, so

$$20x^2 + 26\log(x) \ \leq \ 20x^2 + 26x^2 \ = \ 46x^2.$$

Thus $f(x) \in O(x^2)$ with witnesses $c = 46$, $k = 1$.

2. [8 points] Using the **formal definition of Big-O**, show that $20xlogx$ is $O(26x^2)$ but that $26x^2$ is not $O(20xlog(x))$. You must clearly state what the witnesses to the relationship are to receive full credit.

Solution: Enter your solution here:
Part 1: $20x\log x \in O(26x^2)$ For $x \geq 1$ we have $\log(x) \leq x$, therefore

$$20x\log x \ \leq \ 20x \cdot x \ = \ 20x^2 \ \leq \ 26x^2.$$

Thus $20x \log x \in O(26x^2)$ with witnesses $c = 1$, $k = 1$.

Part 2: $26x^2 \notin O(20x \log x)$ Suppose for contradiction that $26x^2 \in O(20x \log x)$. Then there exist constants $c > 0$ and k such that $26x^2 \leq c \cdot 20x \log x$ for all $x \geq k$, which simplifies to

$$\frac{26x}{20c \log x} \leq 1 \quad \text{for all } x \geq k.$$

However,

$$\lim_{x \to \infty} \frac{26x}{20c \log x} = \frac{1}{c} \lim_{x \to \infty} \frac{26x}{20 \log x} = \infty,$$

a contradiction. Thus $26x^2 \notin O(20x \log x)$.

3. 16 points For each of the following functions, give a proof using the **Limit method** that $f(x) \in O(g(x))$:
   (a) $(f(x) = 20x^2, g(x) = 26x^3$
   (b) $f(x) = 20xlog(x), g(x) = 26x^2$
   (c) $f(x) = 20 + log(x) + 2x^2 + 6x^3, g(x) = x^3$
   (d) $f(x) = 20xlog(x) + 2x, g(x) = 6x^2$

Solution:
Enter your solution here:
A:

$$\lim_{x \to \infty} \frac{20x^2}{26x^3} = \frac{20}{26} \lim_{x \to \infty} \frac{1}{x} = 0.$$

Since the limit is $0 < \infty$, we conclude $f(x) \in O(g(x))$.

B: Divide numerator and denominator by x (valid for $x > 0$):

$$\lim_{x \to \infty} \frac{20x \log x}{26x^2} = \lim_{x \to \infty} \frac{20 \log x}{26x}.$$

Apply L'Hôpital's Rule:

$$\lim_{x \to \infty} \frac{20 \log x}{26x} = \lim_{x \to \infty} \frac{\frac{20}{x}}{26} = \lim_{x \to \infty} \frac{20}{26x} = 0.$$

Since the limit is $0 < \infty$, we conclude $f(x) \in O(g(x))$.

C:

$$\lim_{x \to \infty} \frac{20 + \log x + 2x^2 + 6x^3}{x^3} = \lim_{x \to \infty} \frac{20}{x^3} + \lim_{x \to \infty} \frac{\log x}{x^3} + \lim_{x \to \infty} \frac{2}{x} + \lim_{x \to \infty} 6 = 0 + 0 + 0 + 6 = 6.$$

Since the limit is $6 < \infty$, we conclude $f(x) \in O(g(x))$.

D:

$$\lim_{x \to \infty} \frac{20x \log x + 2x}{6x^2} = \lim_{x \to \infty} \frac{20 \log x}{6x} + \lim_{x \to \infty} \frac{2}{6x}.$$

The second term goes to 0. For the first term, apply L'Hôpital's Rule:

$$\lim_{x \to \infty} \frac{20 \log x}{6x} = \lim_{x \to \infty} \frac{\frac{20}{x}}{6} = \lim_{x \to \infty} \frac{20}{6x} = 0.$$

Therefore,

$$\lim_{x \to \infty} \frac{20x \log x + 2x}{6x^2} \;=\; 0 + 0 \;=\; 0.$$

Since the limit is $0 < \infty$, we conclude $f(x) \in O(g(x))$.

4. [8 points] Answer the following parts:

   (a) Give pseudocode for an algorithm that finds the first repeated negative integer in a given sequence of integers.

   (b) Analyze the worst-case time complexity of the algorithm you devised in part (a). To receive full credit, you must state your choice of elementary operation.

Solution:
A:

```
1 function FirstRepeatedNegative(numbers):
2     seen = empty set
3     for x in numbers:
4     |    if x < 0:
5     |        if x in seen:
6     |            return x
7     |        add x to seen
8     end
9     return "none"
```

B:
Elementary operation: Comparison (lines 3, 4, and 5). With n elements this gives at most $3n$ comparisons. Worst-case time complexity: $O(n)$.

5. [8 points] Answer the following parts:

   (a) Give pseudocode for an algorithm that determines whether a string of n characters has all unique characters.

   (b) Analyze the worst-case time complexity of the algorithm you devised in part (a). To receive full credit, you must state your choice of elementary operation.

Solution:
A:

```
1 function UniqueString(s):
2     seen = empty set
3     for c in s:
4     |    if c in seen:
5     |        return false
6     |    add c to seen
7     end
```

```
8     return true
```

B:
Elementary operation: Comparison (lines 3 and 4). With n elements this gives at most $2n$ comparisons. Worst-case time complexity: $O(n)$.

6. [8 points] Answer the following parts:

   (a) Give pseudocode for an algorithm that locates the last occurrence of the smallest element in a list of n integers. (Elements are not necessarily distinct.)

   (b) Analyze the worst-case time complexity of the algorithm you devised in part (a). To receive full credit, you must state your choice of elementary operation.

Solution:
A:

```
1 function LastMinIndex(ints):
2     min_val = ints[0]
3     index = 0
4     for i from 1 to n-1:
5     |    if ints[i] <= min_val:
6     |        min_val = ints[i]
7     |        index = i
8     end
9     return index
```

B:
Elementary operation: Comparison (lines 4 and 5). The loop always runs from $i = 1$ to $i = n-1$ regardless of the input. Each iteration incurs 2 comparisons: the loop condition on line 4 and the $\leq$ check on line 5, giving $2(n-1)$ comparisons total. Worst-case time complexity: $O(n)$.

7. [8 points] Answer the following parts:

   (a) Given an array of integers $nums$ of length n and an integer target , find three distinct indices i, j, k such that $nums[i] + nums[j] + nums[k] = target$.

   You may assume exactly one such triplet exists, and you may not use the same element more than once. Return the indices in any order.

   (b) Analyze the worst-case time complexity of the algorithm you devised in part (a). To receive full credit, you must state your choice of elementary operation.

Solution:
A:

```
1 function ThreeSum(nums, target):
2     for i from 0 to n-1:
```

```
3     |    for j from i+1 to n-1:
4     |    |    for k from j+1 to n-1:
5     |    |    |    if nums[i] + nums[j] + nums[k] == target:
6     |    |    |        return (i, j, k)
7     |    |    end
8     |    end
9     end
```

B:
Elementary operation: Comparison (lines 2, 3, 4, and 5). The total number of comparisons is given by:

$$\sum_{i=0}^{n-1} \sum_{j=i+1}^{n-1} \sum_{k=j+1}^{n-1} 4$$

Evaluating the innermost sum:

$$\sum_{k=j+1}^{n-1} 4 = 4(n-1-j)$$

Substituting into the middle sum:

$$\sum_{j=i+1}^{n-1} 4(n-1-j) = 4 \sum_{m=0}^{n-2-i} m = 4 \cdot \frac{(n-1-i)(n-2-i)}{2} = 2(n-1-i)(n-2-i)$$

Substituting into the outer sum:

$$\sum_{i=0}^{n-1} 2(n-1-i)(n-2-i) = 2 \cdot \frac{n(n-1)(n-2)}{3} = \frac{2n(n-1)(n-2)}{3} = 4 \cdot \binom{n}{3} \in \Theta(n^3)$$

In the worst case every triple is examined, giving $\frac{2n(n-1)(n-2)}{3}$ total comparisons. Worst-case time complexity: $O(n^3)$.